

Derived Pro-Cokernel and the Recollement Class

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Summary

Pass 70 compares two presentations of the Loeb-Rosser phantom:

1. the derived pro-cokernel from the inverse-limit exact sequence;
2. the recollement extension class ϵ_S from the Loeb-Rosser bicomplex/six-functor package.

For a finite prime set S , define

$$M_{S,k} = \prod_{p \in S} p^k.$$

The levelwise sequence

$$0 \rightarrow M_{S,k}\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/M_{S,k}\mathbb{Z} \rightarrow 0$$

gives

$$\varprojlim_k \mathbb{Z}/M_{S,k}\mathbb{Z}$$

$$\widehat{\mathbb{Z}}_S$$

$$\prod_{p \in S} \mathbb{Z}_p,$$

and therefore

$$\varprojlim^1 (M_{S,k}\mathbb{Z}) \cong \widehat{\mathbb{Z}}_S/\mathbb{Z}.$$

The Comparison

The natural projection from the global derived pro-cokernel to the product of local derived cokernels is

$$\widehat{\mathbb{Z}}_S/\mathbb{Z} \rightarrow \prod_{p \in S} (\mathbb{Z}_p/\mathbb{Z}).$$

Its kernel is made of tuples represented by ordinary integers in each coordinate, modulo the single diagonal integer:

$$\ker = \mathbb{Z}^S/\Delta\mathbb{Z} \cong \mathbb{Z}^{|S|-1}.$$

Hence the extension

$$0 \rightarrow \mathbb{Z}^S/\Delta\mathbb{Z} \rightarrow \widehat{\mathbb{Z}}_S/\mathbb{Z} \rightarrow \prod_{p \in S} (\mathbb{Z}_p/\mathbb{Z}) \rightarrow 0$$

is exactly the recollement class ϵ_S .

Signed Shadow

After choosing a base prime $p_0 \in S$, the boundary matrix is

$$d_S : \mathbb{Z}^S \rightarrow \mathbb{Z}^{|S|-1}, \quad d_S(x) = (x_p - x_{p_0})_{p \neq p_0}.$$

It has kernel $\Delta\mathbb{Z}$ and is surjective. On finite-prime, character-normalized shadows, duality sends

$$d_S \mapsto -d_S^T,$$

so

$$D_{\text{ch}}(\epsilon_S) = -\epsilon_S^\vee.$$

Verification

The checker `code/scripts/check-pass70.py` produces `artifacts/reports/pass70-derived-pro-epsilon-comparison-check.json`.

It verifies for $|S| = 1, \dots, 5$ that:

- CRT finite shadows for $M_{S,k}$, $1 \leq k \leq 4$, are bijective;
- $\Delta : \mathbb{Z} \rightarrow \mathbb{Z}^S$ is primitive;
- $d_S : \mathbb{Z}^S \rightarrow \mathbb{Z}^{|S|-1}$ is surjective;
- $\ker d_S$ is the diagonal by rank;
- $-d_S^T$ double-dualizes back to d_S .

The remaining open problem is not the algebraic comparison. It is the all-prime topological duality normalization: formulate the restricted-product LCA, condensed, or solid category in which the finite signed laws assemble without collapsing products to finite-support direct sums.